HS

Hussain Singapuri

Senior AI Engineer (Agentic AI, Tool Calling and LLM Applications)

3 yrs professional full-stack experience

30+ projects shipped

5 apps live on Google Play

Live portfolio

Contact

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Skills

Core strengths

Agentic AI & Multi-Agent Systems · Tool / Function Calling · Python, FastAPI & asyncio · API Design & Integration · RAG & Vector Search (pgvector) · Agent & LLM Evaluation

AI / LLM

Agentic AI · multi-agent orchestration · tool use / function calling · Claude API · Gemini API · RAG (hybrid BM25 + vector

Back end

Python · FastAPI · asyncio · REST APIs · API design · Node.js

Languages

SQL · TypeScript · JavaScript · Dart · HTML

Data

PostgreSQL · Neon · Supabase (RLS · Edge Functions) · PGlite · pgvector

Education

Bachelor's, Computer Science (AI & Machine Learning)
Vishwakarma Institute of Technology, Pune
2022-2025

Certifications

- Claude Certified Developer
- Claude Certified Orchestrator
- Generative AI Leader
- Claude Code in Action
- Claude Code 101

Profile

AI Engineer with 3 years of professional software experience at a US-based medical-device company, building production agentic AI systems: tool/function calling, multi-step multi-agent orchestration with retries, timeouts and token budgets, RAG with vector databases (pgvector) and hybrid search, eval harnesses and output guardrails, on the Claude API and Gemini API. Strong Python backend skills (FastAPI, asyncio, API design, SQL), Docker and CI/CD on GitHub Actions. Leads technology as CTO at NeuroNetworks AI Ed. Claude Certified Developer.

Experience

CTO & Principal Course Architect — 2026 - Present
NeuroNetworks AI Ed, Pune, India

- Lead technology for an Indian AI and tech education brand (Neuronetworks AI Ed Private Limited) offering programmes for students after Class 10
- Designing Neuromind Practical AI Lab, a browser-based IDE and sandbox with an AI tutor: phased spec, server sizing, engineering team plan and CAPEX/OPEX model
- Architected the curriculum: a 1-year AI Foundation course and five 3-year specialisations (AI/ML Engineering, Software Development, Data Science & AI, Product & UX / AI Design, Cybersecurity)

Full Stack Developer — Mar 2023 - Sep 2026
US-based medical-device company, Remote

- Built AI agents for internal use
- Built and maintained full-stack web applications and several apps, including a CRM
- Performed penetration / security testing on 5+ of the company's mobile apps, all live on the Google Play Store

Selected work

multi-agent-research-crew
Python, asyncio, Claude

multi-agent-research-crew

Agentic AI: planner, parallel researcher, writer and verifier agents on a custom DAG orchestrator with retries, timeouts, token budgets and claim-level citation checking.

Code on GitHub

site-to-agent
Next.js, TypeScript, pgvector

site-to-agent

RAG agent: turns any website into a grounded chat agent with a crawler, hybrid BM25 + vector retrieval (pgvector) fused with RRF, cited answers, SSRF-hardened fetching and an eval harness for answer quality.

Code on GitHub

nanoserve
Python, PyTorch, FastAPI

nanoserve

LLM inference engine from scratch in Python with an OpenAI-compatible streaming FastAPI API: paged KV cache, prefix caching, continuous batching, speculative decoding; 3.1x throughput over sequential decoding in benchmarks.

Code on GitHub

Also built Orbital Surplus, Nightjar & Co., Personal portfolio with full-stack backend, Safey Inventory, TripSync, Lumen, Kilnwood Roasters, Cairnmoor, Lowlight Video Club · github.com/huss2003